

MAT122: Introductory Mathematics II

Comprehensive Tutorial: Week 6 - Differentiation

Department of Mathematics

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1 Plane Curves and Parametric Equations

1.1 Basic Parametric Equations and Sketching

Problem 1: Sketch the curve represented by the parametric equations and indicate the orientation with arrows.

a) $x = t^3, y = t^2, -1 \leq t \leq 2$

b) $x = \cos t, y = \sin t, 0 \leq t \leq 2\pi$

Problem 2: Complete the table of values for the given parametric equations and plot the corresponding points.

t	-2	-1	0	1	2	3
$x = t^2 - 1$						
$y = 2t$						

1.2 Eliminating the Parameter

Problem 3: Eliminate the parameter and write the rectangular equation. Identify the type of curve (line, parabola, circle, etc.).

a) $x = 3t, y = 6t - 2$

b) $x = t^2, y = 2t$

c) $x = 2 \cos t, y = 2 \sin t$

d) $x = e^t, y = e^{2t}$

e) $x = \sqrt{t}, y = t^2, t \geq 0$

Problem 4: For each set of parametric equations, eliminate the parameter and state any restrictions on x or y that come from the parametric domain.

- a) $x = \frac{1}{t}, y = t, t \neq 0$
- b) $x = \ln t, y = t, t > 0$
- c) $x = t^2 + 1, y = \sqrt{t}, t \geq 0$

1.3 Finding Parametric Equations from Rectangular Equations

Problem 5: Find two different sets of parametric equations for each curve.

- a) $y = 2x - 3$
- b) $x^2 + y^2 = 9$
- c) $y = \ln x$

Problem 6: Using slope $m = \frac{dy}{dx}$ as a parameter, find parametric equations for $y = x^3$.

Section 2: Derivatives of Parametric Equations

Problem 7: Find $\frac{dy}{dx}$ for each curve.

- a) $x = 2t, y = t^2 + 1$
- b) $x = \cos t, y = \sin t$
- c) $x = e^t, y = \ln t, t > 0$
- d) $x = t - \sin t, y = 1 - \cos t$

Problem 8: Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for each curve.

- a) $x = \sqrt{t}, y = t^3, t > 0$
- b) $x = 2 \cos 5\theta, y = 2 \sin 5\theta$

Section 3: Slope and Tangent Lines

Problem 9: Find the points on the curve where the tangent is horizontal or vertical.

- a) $x = t^2, y = t^3 - 3t$
- b) $x = \cos t, y = \cos^3 t, 0 \leq t \leq 2\pi$

Problem 10: Find the equation of the tangent line at the given point.

- a) $x = 2t, y = t^2$, at $t = 2$
- b) $x = e^t, y = e^{-t}$, at $t = 0$
- c) $x = \sin t, y = \cos 2t$, at $t = \frac{\pi}{4}$